

SECTION 15849

DUCT HANGERS AND SUPPORTS

PART 1 GENERAL

1.1 SUMMARY

- A. This Section includes hangers and supports for mechanical systems piping, ducts and equipment.
- B. This section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for Duct Hangers and Supports.
 - 1). Division 1 – Submittal Procedures
 - 2). Division 1 – Product Requirements
 - 3). Division 5 – Metal Fabrications.
 - 4). Division 7 – Fire-stopping.
 - 5). Section 15200 – Mechanical Vibration and Noise Control.
 - 6). Section 15200.01_dat – Mechanical Vibration and Noise Control Data Sheets.
 - 7). Section 15811 – Fiberglass Reinforced Plastic (FRP) Ductwork.
 - 8). Section 15813 – Teflon Lined Stainless Steel Exhaust Ductwork and Accessories.
 - 9). Section 15840 – Metal HVAC Ductwork and Accessories (General).
 - 10). Section 15842 – Process Exhaust Ductwork Schedule.
 - 11). Section 15843 – Metal Process Exhaust Ductwork (Specific).
 - 12). Section 230500 – Common Work Results for HVAC.
 - 13). Section 230700 – Equipment, Vessels, Piping, and Duct Insulation.

1.2 DEFINITIONS

- A. Terminology used in this Section is defined in Manufacturers Standardization Society (MSS) SP 90.

1.3 PERFORMANCE REQUIREMENTS

- A. General: Design and provide duct hanger material and appurtenances conforming to the minimum level of acceptance set forth in Manufacturers Standardization Society (MSS) SP 58. This specification indicates general MSS types only and manufacturers' other standard products are acceptable under Standard SP-58, if they meet dimensional and load rating limitation. Select hangers, supports, anchors, guides, and restraints to withstand all static and dynamic condition of loading to which piping and associated equipment may be subjected. Design supplementary structural members for the specific loads they are to support in accordance with the American Institute of Steel Construction (AISC) Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings. Use allowable stress levels listed in Table 2 of MSS SP-58 for Contractor's application of hanger assemblies.

1.4 SUBMITTALS

- A. General: Submit the following according to the Conditions of the Contract and Division 1 Specification Sections.
- B. Product data for each type of hanger and support.
- C. Submit duct hanger and support schedule showing manufacturer's Figure No., size, location, and features for each required hanger and support.
- D. Licensed Engineer: Prepare hanger and support design drawings, and calculations for seismic restraint of piping and equipment. Include seal and signature of Registered Engineer, licensed in jurisdiction where Project is located, certifying compliance with specifications.

1.5 SITE RECORDS

- A. Maintain Site records for welder certificates signed by Contractor certifying that welders comply with requirements specified under the "Quality Assurance" Article.

1.6 QUALITY ASSURANCE

- A. Coordination: Coordinate with the installing contractor for mechanical insulation to establish the insulation material and thickness that will be furnished for each insulated system.
- B. Qualify welding processes and welding operators according to AWS D1.1 "Structural Welding Code—Steel."
- C. Certify that each welder has satisfactorily passed AWS qualification tests for welding processes involved and, if pertinent, has undergone recertification.
- D. Qualify welding processes and welding operators according to ASME "Boiler and Pressure Vessel Code," Section IX, "Welding and Brazing Qualifications."
- E. Comply with referenced codes and manufacturer's standards.
 - 1). ANSI B31.1, Power Piping.
 - 2). ANSI B31.3, Process Piping.
 - 3). ANSI B31.9, Building Services Piping.
 - 4). MSS SP-58, Pipe Hangers and Supports - Materials, Design and Manufacturing.
 - 5). MSS SP-69, Pipe Hanger and Supports Selection and Application.
 - 6). MSS SP-89, Pipe Hangers and Supports Fabrication and Installation Practices.
 - 7). AWS D1.1, Structural Welding Code.
 - 8). SMACNA Standards, latest Edition.
 - 9). International Seismic Application Technology (I.S.A.T).

PART 2 PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Hangers:

- 1). Anvil
 - 2). B-line
 - 3). TOLCO
 - 4). PHD
 - 5). Superstrut
- B. Channel Framing Systems:
- 1). B-Line Systems, Inc.
 - 2). Superstrut
- C. Insulated Pipe Supports:
- 1). Pipe Shields
 - 2). Rilco
 - 3). Valued Engineered Products
- D. Pipe Cushions/Clamps:
- 1). ZSI
 - a). Cush-a-strip
 - b). Cush-a-clamp
 - 2). Cooper B-Line:
 - a). Iso-Pipe Tape
 - b). Armafix
- E. Sleeves:
- 1). Thunderline Corp.
 - 2). Pipe Shields
 - 3). Calpico

2.2 GENERAL REQUIREMENTS

- A. Refer to Section 230500 – Common Work Results for HVAC section 2.5.
- B. Hangers, Supports, and Components: Factory-fabricated according to MSS SP 58.
- 1). Components include galvanized coatings where installed for piping and equipment that will not have a field-applied finish.
 - 2). Pipe attachments include nonmetallic wrapping or coating for electrolytic protection where attachments are in direct contact with copper tubing.

- 3). Hangers shall be sized to fit around bare pipe, isolator, hanger shield or insulation as required.
 - 4). Use FRP hangers in bulk chemical distribution rooms and other highly corrosive areas. Prior to install, verify with the Construction Manager or Engineer of Record which areas are classified as highly corrosive.
- C. Mechanical-Anchor Fasteners: Insert-type attachments with pullout and shear capacities appropriate for supported loads and building materials where used. Size inserts to fit threaded hanger rods. Fasteners for fire protection systems include UL listing and FM approval. Do not use insert-type attachments in lightweight concrete or concrete slabs less than 4 inches thick.
- 1). Inserts for use in poured-in-place concrete MSS Type 18 or equal, malleable iron cases or galvanized steel shells and expander plugs for threaded connection with lateral adjustment, top slot for reinforcing rods and lugs for attaching to forms. Provide manufactured threaded rod inserts for metal decks with concrete, retaining rod to extend 2 inches minimum above top of deck flutes. Inserts may not be used on roof decks without written specific acceptance by the Engineer of Record.
 - 2). Provide ICBO and ASTM E488 tested drilled insert expansion shields, inserts, and fasteners on concrete walls and floors and sides of concrete beams. Fastener devices exposed to corrosive fluids shall be of 316 stainless steel material.
 - 3). Powder-driven fasteners, expansion nail type anchors, or friction spring-type clips will not be permitted.
- D. In areas of metal decking, pipe sizes 4 inch and smaller, and duct shall be supported from beams or joists and intermediate steel with UL/FM MSS Type 23 malleable iron "C" clamps with Anvil Figure No. 89 retainer straps. Provide angles or channels to span between joists or beams as required to maintain support centers specified herein. For loads up to 1,800 pounds, use B Line B3040 Series adjustable beam clamps. For loads greater than 1,800 pounds, use B Line or Grinnell UFS Series forged steel beam clamps. Engineered clamps may be submitted for use with complete calculation package.

2.3 HANGER RODS

- A. Provide steel hanger rods, appropriately threaded. Provide connection points with jamb nuts or double nuts. For highly corrosive environment FRP hanger rods employing vinyl ester resin shall be used.

2.4 INDUSTRIAL DUCT HANGERS AND SUPPORTS

- A. Industrial duct includes all production related systems, including General Exhaust, Fume (Acid) Exhaust, Ammonia (AME) Exhaust, VOC Exhaust, Production Makeup Air and Recirculation.
- B. General:
 - 1). Duct hanger and support material gages shall comply with details provided on the Drawings. If not provided, comply with the latest SMACNA Industrial Duct Construction Standards.
 - 2). Suspension material shall be cadmium plated threaded rod (minimum 3/8 inch diameter) or strut material. The use of wire or metal strap as suspension material is prohibited.

- 3). All ductwork will have vibration isolators installed in accordance with Section 15200 – Mechanical Vibration and Noise Control of these Specifications. Installation of hangers, supports and bracing shall be done in such a manner as not to "short circuit" these isolators. Refer to Section 15200 for vibration isolation requirements.
 - 4). Provide galvanized steel angles, zinc dichromate coated strut channels, stays, rods, etc., for duct support and reinforcement unless painted where specified. FRP supports and hardware shall be used in highly corrosive areas (i.e. bulk chemical distribution rooms).
 - 5). Maximum support spacing on all industrial ductwork is 8 feet.
 - 6). Use of Van Stone angle companion flanges as hanger assembly is prohibited.
- C. Upper Attachments:
- 1). Acceptable Manufacturers:
 - a). Fee and Mason Manufacturing Co.
 - b). Grinnell
 - c). Elcen Metal Products Co.
 - 2). Support ductwork from structural or supplementary steel, utilizing beam clamps designed for the intended service. Support from metal decks above is prohibited. Provide supplementary steel as detailed and as required.
- D. Trapezes: Utilize strut material approved in this section.
- E. Support Frames: Use structural steel shapes or strut frames for support of ductwork accessories, and HVAC components with nuts, bolts, fittings, etc. that are compatible with the frame material.
- F. Metal Ductwork Support:
1. Fabricate and install hangers and supporting systems. Design and install hangers and supporting systems in accordance with Chapter 4 of the SMACNA "HVAC Duct Construction Standards", latest edition; hereinafter referred to in this specification as Duct Manual, with the following exceptions:
 - a) Do not use powder actuated fasteners in pre-stressed concrete.
 - b) Do not drill, cut, or weld building steel.
 - c) Do not connect hangers directly to metal roof decks.
 - d) For ductwork 36" x 30" and larger, and other ductwork of any size, if so located that a man can stand on, sit on or crawl on top of, hangers and support systems shall adequately support ductwork, reinforcement, insulation and trapeze weights plus an additional concentrated load of 500 pounds. Note that Table 4-1 does not allow for this additional load. Information for custom design of hangers for additional loads must be obtained from SMACNA Industrial Design Manual.
 - e) Where vibration isolators are to be installed in the ductwork hangers, use properly sized rods in lieu of straps.
 2. Support Spacing - Maximum Permitted Hanger Spacing:
 - a) Ducts with areas up to 4 square feet may have their hangers spaced up to 8 feet apart.

- b) Ducts with areas from 4 to 10 square feet may have their hangers spaced not more than 6 feet apart.
 - c) Ducts with areas over 10 square feet may have their hangers located up to 4 feet apart.
3. Materials:
- a) Generally, to be constructed of same material as ductwork with which it is in contact.
 - b) If support metal differs from metal used in duct construction, different materials to be separated by a full neoprene gasket to prevent corrosion by electrolysis.
 - c) Hanger rods for trapeze hangers - galvanized or cadmium plated steel, unless noted otherwise.
 - d) Provide steel hanger rods, appropriately threaded. Provide connection points with jamb nuts or double nuts.
 - e) For highly corrosive environment FRP hanger rods employing vinyl ester resin shall be used.
4. Upper attachment system for each duct hanger shall be sized with a safety factor of 4 for the maximum allowable load for the duct hanger itself. These values are in accordance with Figure 4-2 and 4-3 of the Duct Manual. Where sheet metal strap hangers are used, connection between the upper attachment device and strap hanger shall be made with a minimum of two 1/4" diameter bolts for 1" hangers and a minimum of two 3/8" diameter bolts for 1-1/2" hangers. Bolted connections are to be made in series down the hanger, not side by side.
5. Fasteners:
- a) Galvanized Steel Ductwork - cadmium plated or galvanized steel screws or 3/16" dia. steel pop rivets.
6. Use the following table or the load limits published by the hanger components manufacturer whichever is more stringent.

<u>Max. Total Load Per Rod</u> ⁽¹⁾	<u>Min. Hanger Rod</u>
610 lb.	3/8 in.
1130 lb.	1/2 in.
1810 lb.	5/8 in.
2710 lb.	3/4 in.
3770 lb.	7/8 in.
4960 lb.	1 in.
8000 lb.	1-1/4 in.
11,630 lb.	1-1/2 in.
15,700 lb.	1-3/4 in.
20,700 lb.	2 in.

⁽¹⁾including trapeze, attachments, insulation

2.6 NON-INDUSTRIAL DUCT HANGERS AND SUPPORTS

- A. Non-industrial includes office, conference room, janitor rooms, non-clean corridors, and break room supply, return, and exhaust ductwork.
- B. General:

- 1). Material Gages: Comply with Drawing details and the latest edition of SMACNA HVAC Duct Construction Standards, Metal and Flexible.
 - 2). Refer to Section 15200 – Mechanical Vibration and Noise Control for ductwork vibration isolation requirements.
- C. Hangers: Threaded rod, metal strap, or strut material is acceptable. Use of wire is not acceptable. Use of companion flange as part of hanger assembly is not acceptable.
- D. Upper Attachments:
- 1). Concrete Construction: Cast-in-Place inserts and concrete expansion anchors, as specified in other division work, are acceptable. Powder actuated fasteners are not acceptable without prior written acceptable written acceptance by the Engineer of Record.
 - 2). Steel Construction: Beam clamps designed for the intended service with supplementary steel s detailed or required elsewhere are acceptable. Support from metal decks is not acceptable.
 - a). Acceptable Manufacturers:
 1. Fee and Mason Manufacturing Co.
 2. Grinnell.
- E. Trapeze:
- 1). Material: Cadmium or Zinc coated galvanized steel strut material.
 - 2). Dissimilar Material Separation: 1/16 inch rubber or neoprene sheet.
- F. Support Frames:
- 1). Material: Structural steel shapes and strut frames complete with nuts, bolts, fittings, etc. compatible with the frame material.

2.7 FLASHING

- A. Steel Flashing: 26 gage galvanized steel.
- B. Lead Flashing: 4 lb/sq ft sheet lead for waterproofing; 1 lb/sq ft sheet lead for soundproofing.
- C. Safes: 5 lb/sq ft sheet lead or 8 mil thick neoprene.
- D. Caps: Steel, 22 gage minimum, 16 gage at fire-resistant structures.

2.8 SLEEVES

- A. Sleeves for Pipes Passing Through Non-rated, Non-bearing Walls and Floors: 20 gage, galvanized sheet metal tube with welded longitudinal joints.
- B. Sleeves for Pipes Passing Through Beams and Footings: Schedule 40 black steel pipe. For waterproof sleeves, use Thunderline Link-Seal or Calpico Sealing Linx.
- C. Penetration through Fire-Rated Walls and Floor: Coordinate the Work of this Section with the requirements of Section 15063 – Firewall Penetration Criteria.

- D. Sleeves for Ducts: Form with minimum 20 gage galvanized steel.
- E. Size sleeves large enough to allow for movement due to expansion and to provide for continuous insulation.

2.9 MISCELLANEOUS MATERIALS

- A. Structural Steel: ASTM A36, steel plates, shapes, and bars, black and galvanized, ASTM A123.
- B. Bolts and Nuts: ASME B18.10 or ASTM A183, steel, hex-head, track bolts and nuts.
- C. Washers: ASTM F844, steel, plain, flat washers.
- D. Grout: ASTM C1107, Grade B, non-shrink, nonmetallic.
 - 1). Characteristics include post-hardening, volume-adjusting, dry, hydraulic-cement-type grout that is non-staining, non-corrosive, nongaseous and is recommended for both interior and exterior applications.
 - 2). Design Mix: 5000 psi, 28-day compressive strength.
 - 3). Water: Potable.
 - 4). Packaging: Premixed and factory-packaged.
- E. Trapeze: Channel ASTM A570, Grade 33, Hot Rolled Structural Quality; finish ASTM B633.
- F. Channels: Framing Channels: 12-gage channel. Unless required otherwise for a specific application, channels are to be zinc dichromate dip coated after electro-galvanized fabrication. Outdoor where coating is specified by Engineer of Record and Construction Manager, channels are to be powder coated per manufacturer's specification, color is to match application.

PART 3 EXECUTION

3.1 INSERTS AND EXPANSION SHIELDS

- A. Use inserts for suspending hangers from reinforced concrete slabs and sides of reinforced concrete beams where continuous embedded insert type channel has not been provided.
- B. Set inserts for poured-in-place-concrete in position in advance of concrete work. Provide reinforcement rod in concrete for inserts carrying pipe over 4 inches in diameter or ducts over 60 inches wide.
- C. Where concrete slabs form finished ceiling, finish inserts flush with slab surface.
- D. Locate expansion shields in concrete beams a minimum of 6 inches above bottom of beam.
- E. Do not use power-driven fasteners or friction spring-type clips.

3.2 HANGER AND SUPPORT APPLICATIONS

- A. General hanger selections and requirements are specified in this Section.
- B. Comply with MSS SP-69 for hanger selections and applications that are not specified in this Section or in process piping specification Sections.

- C. Coordinate the Work of this Section with the requirements of Section 15200 – Mechanical Vibration and Noise Control.
- D. Coordinate the Work of this Section with I.S.A.T. pre-approval R-0485 for materials applications.
- E. Do not use wire for either temporary or permanent hangers or supports.

3.3 HANGERS AND SUPPORTS INSTALLATION

- A. Duct hangers, supports, and installation per latest edition SMACNA Standards. All minor components such as straps, threaded rods, nuts, and washers shall be galvanized or cadmium plated. FRP components shall be used in highly corrosive areas (i.e. bulk chemical distribution rooms).
- B. Specific details shown on Drawings take precedence over latest edition SMACNA requirements.
- C. General Support Locations:
 - 1). Main Duct and Branches: Within 6" of transverse joints, clinch collar branch connections, and the first branch elbows after nested splits.
 - 2). Wall Penetrations: Either side of wall, allowing no wall support of the duct.
 - 3). Wall Supports: Where required.
 - 4). Vaned Elbows: At midpoint when width is greater than 30 inches, in addition to the maximum allowable intervals measured along the heel line.
 - 5). Radius Elbows: At midpoint when greater than 20 degree offset or whenever the length of arc exceeds the maximum hanger spacing in length.
 - 6). Short Branch Ducts: Greater than 12" in Length.
 - 7). Vertical Ducts:
 - a). Single duct support – minimum of 2 supports completely spanning the shaft opening at each floor.
 - b). Multiple duct supports – increase the support strength and add supplementary steel when supporting more than 1 duct.
 - c). Between floors – minimum of 1 mid-floor support with 12 feet maximum for round ducts and 10 feet for rectangular ducts.
 - d). Intermediate supports – attached to adjacent wall or suspended from the floor above.
- D. Specific Support Locations:
 - 1). Fan Flex Connection: 2 to 24 inches.
 - 2). Terminal Unit or Mixing Box: 2 to 24 inches from outlet or flex connection.
 - 3). Main to Branch Duct Connection: 12 to 36 inches from main duct.
 - 4). End of Branch Duct: 2 to 12 inches.

- 5). Fire/Smoke Damper: 2 to 24 inches from breakaway points.
- 6). Vertical Offsets: Vaned Rectangular Elbows:
 - a). Offset shorter than 8 feet 2 inches to half the duct width and 2 inches from the vertical with heavier duct hangers closer to centerline.
 - b). Offsets greater than or equal to 8 feet 2 inches to half the duct width plus 2 inches from the vertical with heavier duct hangers closer to centerline.
- 7). Vertical Offsets: Radiused Elbows:
 - a). Offsets less than 8 feet at the bottom of the elbow.
 - b). Offsets greater than or equal to 8 feet, 1/8 of the arc from the ends of bottom and top radius elbows.
- 8). Wall Penetrations: 6 to 12 inches from one side.
- 9). Provide supports at fittings and takeoffs as required to prevent stresses at joints.
- E. Support Spacing: Maximum 8 feet O.C. for Industrial ductwork and per latest edition of SMACNA for Non-Industrial ductwork.
- F. Support Method:
 - 1). Low Ductwork: Strut or other structural shapes from floors or other steel structures that are acceptable to the Engineer of Record. Maintain 6 inches minimum clearance between the bottom of duct and the supporting structure.
 - 2). High Ductwork: Overhead using threaded rod attached to trapeze supports for rectangular duct, or circumferential straps for round ducts.
 - 3). Support Material: Trim to minimum length after adjusting elevation.
 - 4). Do not use sheet metal screws to attach support straps to Industrial ducts.
 - 5). Support off structural steel, supplementary steel or concrete ribs. Support off metal decks above is prohibited. Provide supplementary steel supports as needed.

3.4 EQUIPMENT SUPPORTS

- A. Fabricate structural steel stands to suspend equipment from structure above or support equipment above floor.
- B. Grouting: Place grout under supports for equipment and pipe stands, and make a smooth bearing surface.

3.5 METAL FABRICATION

- A. Cut, drill, and fit miscellaneous metal fabrications for pipe and equipment supports.
- B. Fit exposed connections together to form hairline joints. Field-weld connections that cannot be shop-welded because of shipping size limitations.
- C. Field Welding: Comply with AWS D1.1 procedures for manual shielded metal-arc welding, appearance and quality of welds, methods used in correcting welding work, and the following:

- 1). Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
- 2). Obtain fusion without undercut or overlap.
- 3). Remove welding flux immediately.
- 4). Finish welds at exposed connections so that no roughness shows after finishing, and so that contours of welded surfaces match adjacent contours.

3.6 PRIMING

- A. Prime coat exposed steel hangers and supports that are not galvanized, cadmium plated, or factory painted. Hangers and supports located in crawl spaces, pipes shafts and suspended ceiling spaces are considered exposed.

3.7 FLASHING

- A. Flash and counter-flash where mechanical equipment passes through weather or waterproofed walls, floors, and roofs.
- B. Flash vent and soil pipes with 24 inches by 24 inches sheet lead, minimum 8 inches above roof. Counter flash with caulked stack flashing sleeve, Zurn Z 196.
- C. Flash floor drains over finished areas and roof drains with lead, 10 inches clear on sides, minimum 36 inches by 36 inches sheet size. Fasten flashing to drain with clamping device.
- D. Provide 8 inches minimum height curbs for roof-mounted mechanical equipment. Flash and counter flash with galvanized steel, soldered and waterproofed.
- E. Provide continuous lead or neoprene safes below air supply unit casing, built-up mop sinks, shower stalls, shower room floors located above rooms. Solder at joints, flash into floor drains, and turn up 6 inches into walls or to top of curbs, and caulk into joints.

3.8 SLEEVES

- A. Set sleeves in position in advance of concrete work. Provide suitable reinforcing around sleeves.
- B. Where piping or ductwork passes through floor, ceiling or wall, close off space between pipe or duct and construction with noncombustible insulation. Provide tight-fitting metal caps on both sides and caulk.
- C. Install chrome-plated escutcheons where piping passes through finished surfaces.
- D. Provide pipe sleeves for all mechanical piping. Coordinate requirements with Section 07840 – Fire-stopping.
- E. Adequately sleeve pipe passing through concrete or masonry walls or concrete slabs to receive both pipe and insulation pertaining thereto.
- F. Waterproof sleeves shall be Thunderline Link-Seal or Calpico Sealing Linx.
- G. Seal pipes passing through walls or slabs. Use mastic or oakum seal in the annular space in non-fire-rated walls.

- H. Insulated pipe shall be insulated in sleeves, sealed, and pointed as above.
- I. Install rated sleeves on pipes as they are being hung, ready for proper placement in rated walls as walls are being constructed.
- J. Where sleeves have been inadvertently omitted in concrete floors, provide requisite pipe opening by using properly sized diamond core drills. Areas located below drilling operations shall be protected from possible damage.

3.9 PENETRATION SEALING (FIRE-RATED)

- A. Coordinate the Work of this Section with the requirements of Section 07840 – Fire-stopping.

3.10 ADJUSTING

- A. Hanger Adjustment:
 - 1). Adjust hangers to distribute loads equally on attachments and to achieve indicated slope of pipe.
 - 2). Coordinate adjustment with requirements of Section 15200 – Mechanical Vibration and Noise Control.

3.11 PAINTING

- A. Touching Up: Clean field welds and abraded areas of shop paint and paint exposed areas immediately after erection of hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.
- B. Apply by brush or spray to provide a minimum dry film thickness of 2.0 mils.
- C. Touching Up: Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal is specified in Section 09900 - Paints and Coatings and Section 09962 – Chemical Resistant Coating.
- D. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A780.

END OF SECTION